Healthcare Insurance Claim Analysis

Python + Power BI | 15,000 members | Claim cost drivers and risk tiers

VICKY JEERU
DATA ANALYST | POWER BI DEVELOPER
Hyderabad, India
jeeruvicky51@gmail.com
+91 91541 39298
LinkedIn **Portfolio** **GitHub**

$201M
total claims analyzed across 15,000 members

3.7x
higher average claim for smokers ($32.3K vs $8.7K)

47.8%
of all claim dollars come from the 19.8% of members who smoke

31.8%
of claim dollars come from the costliest 10% of members (97% are smokers)

Executive Summary

Health insurers must price policies and plan reserves around who will generate costly claims. I analyzed **15,000 member records** in Python (Pandas and exploratory data analysis) to find the factors that drive claim amounts, segmented members into simple **risk tiers**, and published the results in an interactive **Power BI dashboard**.

> **Bottom line:** smoking status dominates everything else. Smokers are 19.8% of members but generate **47.8% of claim costs**, and **smokers who are also obese average $41,086 per claim, 4.7x the $8,731 average for non-smokers**. Age, BMI and diabetes matter mainly because they *amplify* the smoking effect.

Key Findings at a Glance

- **Smoking is the #1 cost driver** (correlation with claim 0.77, versus 0.30 for age and 0.20 for BMI). Average claim: $32.3K for smokers vs $8.7K for non-smokers.
- **BMI amplifies smoking risk:** obese smokers average $41.1K vs $22.6K for smokers of normal weight (+82%), while obese non-smokers are only 26% above normal-weight non-smokers.
- **Age raises claims steadily:** among non-smokers, members aged 50-64 average $13.5K vs $4.7K for ages 18-29 (2.9x).
- **Diabetes matters most for smokers:** +52% for smokers ($34.3K vs $22.6K) but only +10% for non-smokers.
- **Costs are highly concentrated:** the costliest 10% of members generate 31.8% of claim dollars, and 97% of them are smokers. Two columns (smoking, BMI) are enough to build practical risk tiers.

Business Problem

Insurance pricing that treats every member alike overcharges healthy members and underprices high-risk ones. The questions: **which member characteristics drive claim costs, how concentrated are those costs, and how should that change pricing and prevention programs?**

Tools & Techniques

Area	What I used
Analysis	Python: Pandas, NumPy, Seaborn, Matplotlib; correlation analysis and group-level EDA
Dashboard	Power BI: KPI cards, slicers and 6 chart visuals on a one-page interactive report
Dataset	15,000 member records, 13 original attributes (demographics, BMI, blood pressure, diabetes, smoking, exercise, hereditary disease, claim amount)

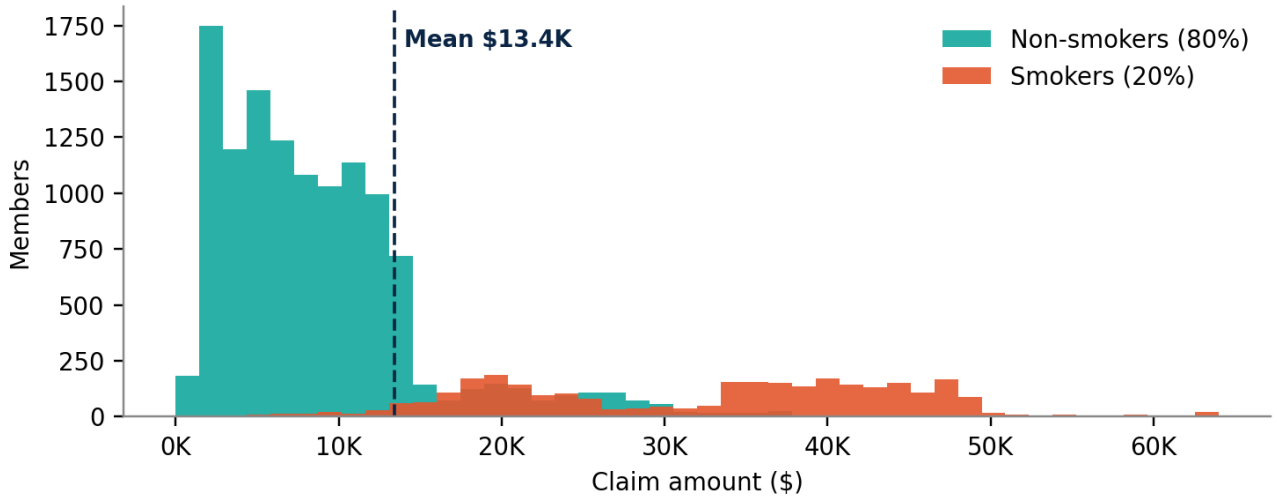
Vicky Jeeru | Data Analyst | jeeruvicky51@gmail.com | +91 91541 39298

1. Data & Preparation

The raw file has 15,000 records. I profiled each column, handled missing values, and engineered category features used by both the analysis and the dashboard.

Step	What I did	Why it matters
1. Profile	Reviewed types, ranges, nulls and outliers (boxplots for age and BMI)	Claims are right-skewed: mean $13,401 vs median $9,546, max $63,770
2. Missing values	Age: 396 missing (2.6%). BMI: 956 missing (6.4%). Filled each with its own **median**	Median is robust to the outliers seen in both columns
3. Imputation audit	Found that a chained fill had set the 956 missing BMI values to 40.0 (the age median). Corrected them to the BMI median (29.4) for this report	Left uncorrected, it creates a false spike of 971 members at BMI 40.0 and inflates the Obese group by 971 members
4. Feature engineering	Built BMI category (WHO bands), age category and blood-pressure category	Turns continuous values into business-readable segments for the dashboard
5. Data quality flags	756 records (5%) have blood pressure of 0; 1,096 rows are exact duplicates; BP categories put 99.8% of members in "Normal"	BP is excluded from segmentation. Duplicates are flagged but kept: removing them leaves results unchanged (smokers: 20.1% of members, 48.0% of claims)

How claim amounts are distributed



Why this matters: the claim distribution has two humps. The right-hand hump is almost entirely smokers, which is the first sign that one variable splits the portfolio into two very different populations. A single "average claim" would mislead pricing.

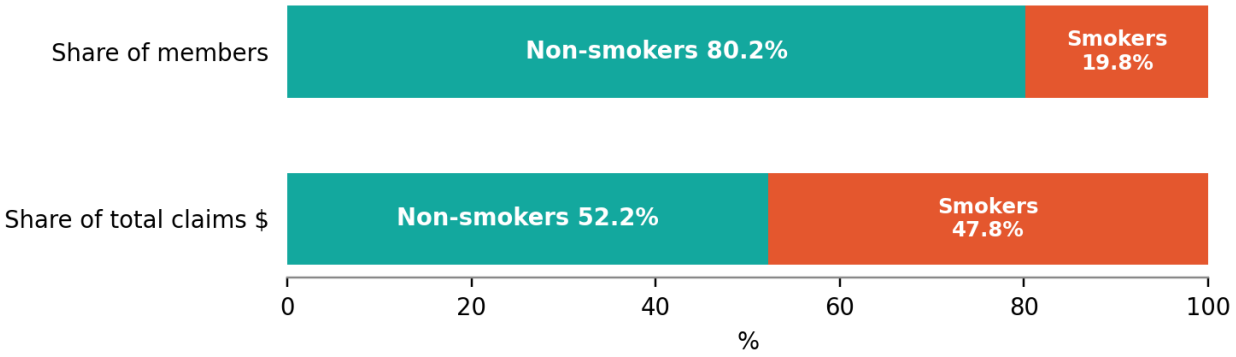
Dataset profile

Attribute	Summary
Members	15,000 (51% female, 49% male); ages 18-64, average 39.6
Smokers / diabetes / regular exercise	19.8% / 77.7% / 22.4% of members
Average BMI (after correction)	30.2; 42.5% of members are obese (BMI 30+)
Claim amount	Average $13,401; median $9,546; 25th-75th percentile $4,847-$16,519

2. What Drives Claim Costs?

Finding 1: Smoking status dominates

Smokers are a fifth of the portfolio but pay out nearly half of all claim dollars. A Mann-Whitney test confirms the difference is overwhelming ($p < 0.001$), and 97% of the 1,500 highest-cost members are smokers.

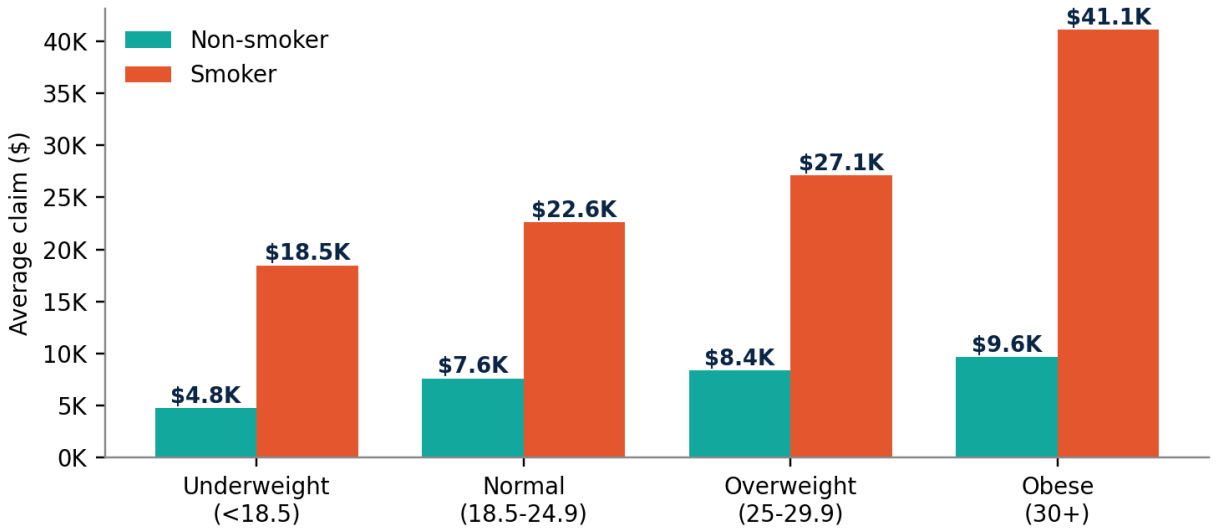


Group	Members	Average claim	Median claim	Share of claim $
Non-smokers	12,028	$8,731	$7,512	52.2%
Smokers	2,972	$32,302	$34,839	47.8%

Business insight: a 19.8% share of members drives 47.8% of cost. Any pricing, underwriting or wellness strategy that ignores smoking status is leaving the single largest cost lever untouched.

Finding 2: BMI raises risk, and smoking multiplies it

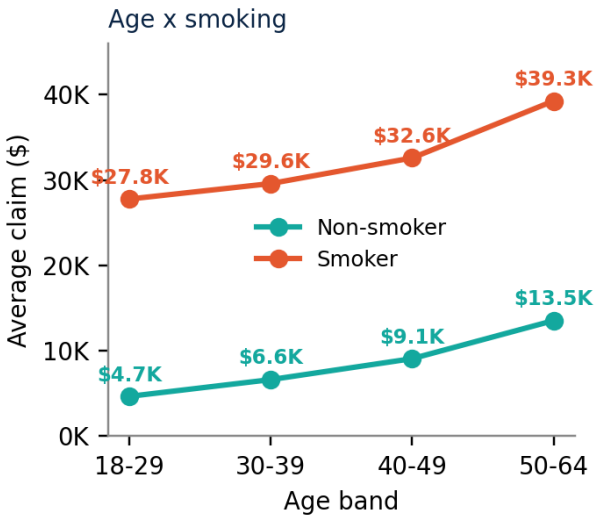
Across all members, obese individuals average $16.0K vs $10.7K for normal weight. The real story appears when BMI is split by smoking status:



Obese smokers average **$41,086**, which is **5.4x** the claim of normal-weight non-smokers ($7,616). Weight status alone moves non-smokers by only about $2K, but moves smokers by about $18K.

3. Age, Diabetes and Lifestyle

Finding 3: Age increases claims, with a smoking premium at every age

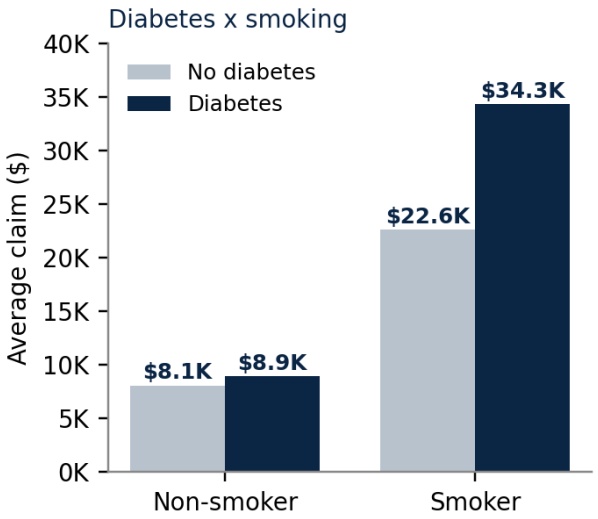


Average claims rise with age in both groups. Among non-smokers they climb from **$4.7K at 18-29 to $13.5K at 50-64** (2.9x). Smokers start far higher ($27.8K at 18-29) and reach $39.3K at 50-64.

- **A young smoker costs more than a 50-64 year-old non-smoker** ($27.8K vs $13.5K)
- Age correlates with claims at 0.30, the second-strongest linear signal after smoking

This is the strongest argument for pricing on smoking status before age.

Finding 4: Diabetes is expensive mainly for smokers



With 77.7% of members flagged diabetic, the average hides a split. Diabetic **non-smokers cost only 10% more** ($8.9K vs $8.1K), while diabetic **smokers cost 52% more** ($34.3K vs $22.6K).

- Diabetes has a modest direct effect and a large interaction with smoking
- Prevention programs should prioritize diabetic smokers first

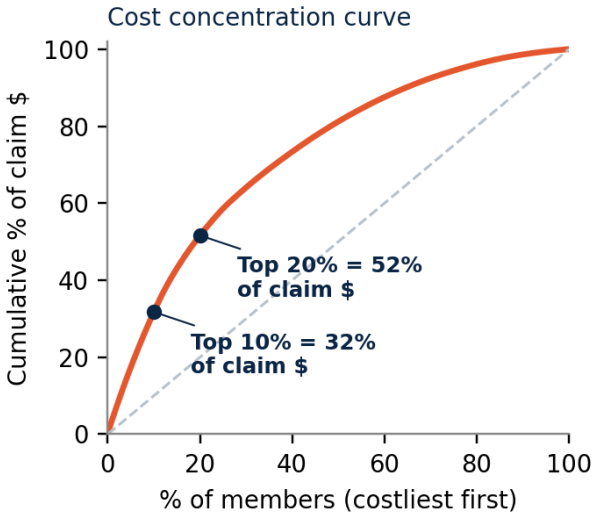
Other signals

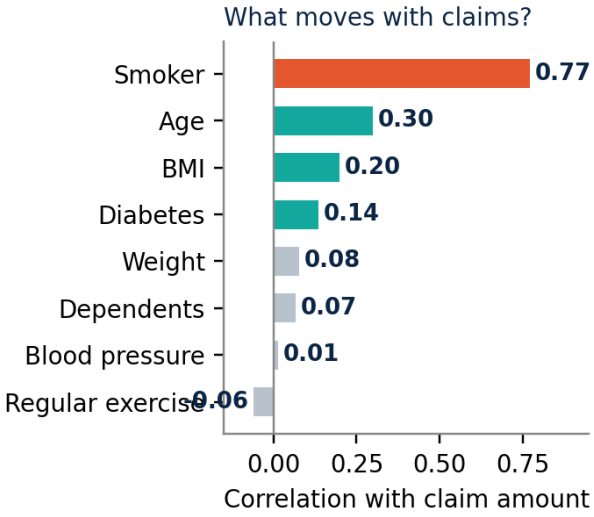
Factor	What the data shows	Takeaway
Regular exercise	Smokers who exercise average $24.5K vs $34.2K for those who do not (-28%, n = 574). No clear effect for non-smokers	Promising for smoker wellness programs; observational, so treat as a hypothesis
Sex	Men average $14.1K vs $12.7K for women (+11%), but men smoke more (22.8% vs 16.9%). Within smoking groups the gap is small	Apparent sex difference is largely a smoking-mix effect
Hereditary disease	1,002 members (6.7%) with a hereditary condition average $39.4K vs $11.5K, but 93% of them smoke	Cannot separate genetics from smoking in this data

Analytical honesty: sex and hereditary disease look like drivers until smoking is controlled for. Blood pressure showed no signal (correlation 0.01).

4. Cost Concentration & Risk Segmentation

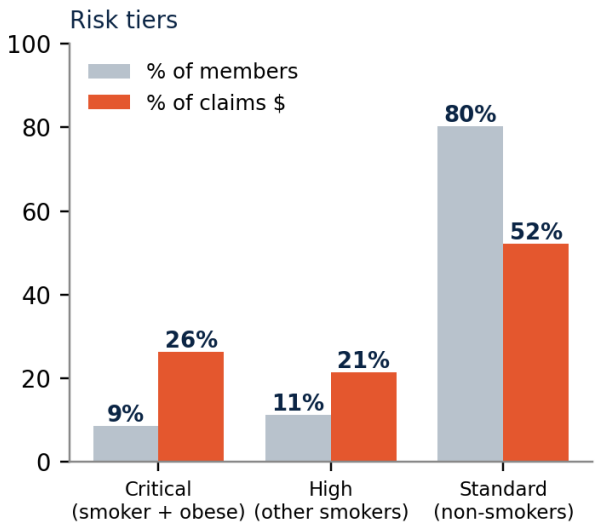
Finding 5: A small share of members drives a large share of cost





- The costliest **5% of members generate 17.4%** of claim dollars, the top **10% generate 31.8%** and the top **20% generate 51.6%**.
- **97.2% of the costliest 10% are smokers.** Smoking status is also the only strong correlate of claims (0.77); age (0.30) and BMI (0.20) are moderate, and blood pressure (0.01) is negligible.

Finding 6: Two columns are enough to build practical risk tiers



Using only smoking status and BMI, I assigned every member to a tier with a single `np.select` rule in Pandas:

- **Critical** (smoker, BMI 30+): 8.6% of members, 26.3% of claim $, average **$41.1K**
- **High** (other smokers): 11.2% of members, 21.4% of claim $, average **$25.6K**
- **Standard** (non-smokers): 80.2% of members, 52.2% of claim $, average **$8.7K**

The Critical tier averages **4.7x** the Standard tier. The tier column is exported to Power BI as a slicer-ready field.

Sample Pandas logic

```
conditions = [(df["smoker"] == 1) & (df["bmi"] >= 30),
              (df["smoker"] == 1)]
df["risk_tier"] = np.select(conditions,
    ["1 - Critical (smoker, BMI 30+)", "2 - High (other smokers)"],
    default="3 - Standard (non-smokers)")
df.groupby("risk_tier")["claim"].agg(["count", "mean", "sum"])
```

Analyst toolkit demonstrated

Skill	Where it was used
Pandas data cleaning	Median imputation per column, duplicate and invalid-value checks, imputation audit
Feature engineering	`pd.cut` age bands, BMI categories, `np.select` risk tiers

Skill	Where it was used
Group-level EDA	`groupby`, `pivot_table` and multi-factor comparisons (smoking x BMI, age, diabetes, exercise)
Statistics	Correlation analysis, medians vs means for skewed data, Mann-Whitney test, confounding checks
Visualization	Seaborn and Matplotlib charts; Power BI dashboard with slicers

5. Dashboard, Recommendations & Next Steps

Power BI dashboard

The findings are published as a one-page interactive report (16 visuals) built on the cleaned dataset exported from Python. Stakeholders can filter every chart at once by smoking, diabetes, sex and exercise.

Component	Purpose
KPI cards	Headline claim totals and member health metrics for the current filter
Claims by age category, BMI category, diabetes, smoker, sex, hereditary disease	Six views showing where claim dollars concentrate (column, ribbon, pie, donut, bar and custom visual)
Slicers: smoker, diabetes, sex, regular exercise	Let a pricing or wellness team test any segment in seconds

Recommendations

#	Action	Evidence
1	**Price on smoking status first,** with an additional loading for obese smokers	Smokers: 19.8% of members, 47.8% of claim $; 3.7x average claim
2	**Target cessation and weight-management programs** at the Critical tier	Critical tier: 8.6% of members, 26.3% of claim $, $41.1K average
3	**Prioritize diabetic smokers** for care management	Diabetes adds +52% for smokers but only +10% for non-smokers
4	**Pilot exercise incentives** for smokers	Exercising smokers average 28% lower claims (observational, n = 574)

Limitations

- Observational data: patterns show association, not causation; claims may reflect pricing or coverage design.
- Blood pressure contains invalid zeros, and the dataset has no claim dates, so trends over time cannot be studied.
- Hereditary disease and smoking overlap heavily (93%), so their separate effects cannot be isolated.

Next steps

- Add DAX measures (average claim per member, smoker share, tier share) and a star schema to the dashboard.
- Add a risk-tier slicer and a cost-concentration visual to the Power BI report.
- Collect claim dates to analyze seasonality and cost trends over time.

Vicky Jeeru | Data Analyst & Power BI Developer | Hyderabad, India
jeeruvicky51@gmail.com | +91 91541 39298 | GitHub | Portfolio | LinkedIn
Code and dashboard: https://github.com/jeeruvicky51-bit/Healthcare-Insurance-Claim-Analysis